

BANK LOAN PORTFOLIO — ADVANCED EXCEL ANALYSIS

PURPOSE

This workbook demonstrates advanced Excel formula techniques applied to the same 38,576-loan portfolio used in the SQL (SQLite) and Python sections of this project.

It shows how an analyst can reproduce and extend key KPIs, risk classification, and loan economics using formulas alone, directly inside Excel.

SHEETS IN THIS WORKBOOK

1. Raw Data - Source loan-level data (38,576 rows) feeding every formula in this workbook.
2. KPI Dashboard - Portfolio KPIs built with SUMIFS / COUNTIFS / AVERAGEIFS / MAXIFS / MINIFS.
3. Risk Classification - Nested IFS + INDEX-MATCH lookup tables to grade DTI and Rate risk bands.
4. Loan Amortization - PMT / IPMT / PPMT / NPER / CUMIPMT functions for a single sample loan.
5. Advanced Lookups - INDEX-MATCH multi-criteria lookups, SUMPRODUCT, TEXTJOIN, IFERROR patterns.

CONVENTIONS

Blue text on yellow fill = editable input cell. Black text = live formula.

All figures recalculate automatically if the Raw Data sheet is refreshed with a new data extract.

PORTFOLIO KPI DASHBOARD (Formula-Driven)

GOOD LOAN vs BAD LOAN

Category	Applications	Funded Amount	Amount Received	Avg Interest Rate
Good Loan (Fully Paid/Current)	33243	\$37,02,24,850	\$43,57,86,170	13.37%
Bad Loan (Charged Off)	5333	\$6,55,32,225	\$3,72,84,763	13.88%
Bad Loan % of Book	13.82%			

TOP STATES BY FUNDED AMOUNT

State	Funded Amount
CA	\$7,84,84,125
NY	\$4,20,77,050
TX	\$3,12,36,650
FL	\$3,00,46,125
NJ	\$2,16,57,475
IL	\$1,71,24,225
VA	\$1,59,82,650
PA	\$1,58,26,525
GA	\$1,54,80,325
MA	\$1,50,51,000

PORTFOLIO EXTREMES

Largest Fully-Paid Loan (\$)	\$35,000
Smallest Charged-Off Loan (\$)	\$900
Highest Interest Rate — Grade A	9.63%
Highest Interest Rate — Grade G	24.59%
Applications — Small Business Purpose	1776
Charge-Off Rate — Small Business Purpose	25.62%

BORROWER RISK CLASSIFICATION (Lookup-Table Driven)

DTI Risk Band Table

DTI Lower Bound	Band
0.00%	Low
15.00%	Medium
25.00%	High

Interest Rate Risk Band Table

Rate Lower Bound	Band
0.00%	Low
12.00%	Medium
18.00%	High

Overall Risk Tier Matrix

DTI Band \ Rate Band	Low	Medium	High
Low	Low Risk	Low Risk	Medium Risk
Medium	Low Risk	Medium Risk	High Risk
High	Medium Risk	High Risk	High Risk

Sample Borrowers

ID	DTI	Interest Rate	DTI Band	Rate Band	Overall Risk Tier
1077430	1.00%	15.27%	Low	Medium	Low Risk
1072053	5.35%	18.64%	Low	High	Medium Risk
1069243	20.88%	15.96%	Medium	Medium	Medium Risk
1041756	5.40%	10.65%	Low	Low	Low Risk
1068350	2.31%	6.03%	Low	Low	Low Risk
1062608	12.60%	14.65%	Low	Medium	Low Risk
1067441	24.53%	14.27%	Medium	Medium	Medium Risk
1066424	6.65%	7.90%	Low	Low	Low Risk
1065254	3.94%	8.90%	Low	Low	Low Risk
1064589	17.73%	12.69%	Medium	Medium	Medium Risk
1057766	8.80%	12.69%	Low	Medium	Low Risk
1062734	22.88%	12.42%	Medium	Medium	Medium Risk
1062654	4.01%	11.71%	Low	Low	Low Risk
1020855	15.33%	12.69%	Medium	Medium	Medium Risk
1060945	10.60%	12.42%	Low	Medium	Low Risk
1060995	6.30%	12.42%	Low	Medium	Low Risk
1046507	3.50%	9.91%	Low	Low	Low Risk
1059936	18.69%	14.27%	Medium	Medium	Medium Risk
1059497	13.92%	10.65%	Low	Low	Low Risk
1058060	9.57%	16.29%	Low	Medium	Low Risk
112245	2.55%	7.43%	Low	Low	Low Risk
207910	7.67%	7.43%	Low	Low	Low Risk
65426	11.08%	11.14%	Low	Low	Low Risk
211723	5.12%	11.09%	Low	Low	Low Risk
662350	7.95%	7.66%	Low	Low	Low Risk

LOAN AMORTIZATION CALCULATOR

Loan Amount (\$)	\$10,000
Annual Interest Rate	11.76%
Term (months)	36

Monthly Payment (PMT)	\$331
Total Repaid Over Term	\$11,916
Total Interest Paid	\$1,916
Cumulative Interest, First 12 Months	\$1,020

AMORTIZATION SCHEDULE

Month	Payment (PMT)	Principal (PPMT)	Interest (IPMT)	Remaining Balance
0	\$331			
1	\$331	\$233	\$98	
2	\$331	\$235	\$96	
3	\$331	\$238	\$93	
4	\$331	\$240	\$91	
5	\$331	\$242	\$89	
6	\$331	\$245	\$86	
7	\$331	\$247	\$84	
8	\$331	\$249	\$82	
9	\$331	\$252	\$79	
10	\$331	\$254	\$77	
11	\$331	\$257	\$74	
12	\$331	\$259	\$72	
13	\$331	\$262	\$69	
14	\$331	\$264	\$67	
15	\$331	\$267	\$64	
16	\$331	\$270	\$61	
17	\$331	\$272	\$59	
18	\$331	\$275	\$56	
19	\$331	\$278	\$53	
20	\$331	\$280	\$51	
21	\$331	\$283	\$48	
22	\$331	\$286	\$45	
23	\$331	\$289	\$42	
24	\$331	\$292	\$39	
25	\$331	\$294	\$37	
26	\$331	\$297	\$34	
27	\$331	\$300	\$31	
28	\$331	\$303	\$28	
29	\$331	\$306	\$25	
30	\$331	\$309	\$22	
31	\$331	\$312	\$19	
32	\$331	\$315	\$16	
33	\$331	\$318	\$13	

34	\$331	\$321	\$10
35	\$331	\$325	\$6
36	\$331	\$328	\$3

ADVANCED LOOKUP & AGGREGATION TECHNIQUES

1) MULTI-CRITERIA LOOKUP

Search State	CA
Search Purpose	small business
First Matching Loan ID	No match found
Funded Amount of That Loan	No match found

2) SUMPRODUCT — LOAN-WEIGHTED AVG RATE, CHARGED-OFF LOANS

Loan-Weighted Avg Rate (Charged Off)	14.71%
(vs. simple AVERAGEIFS, for comparison)	13.88%

3) TEXTJOIN — READABLE STATE SUMMARY LABELS

State	Summary Label
CA	CA: 6894 loans \$7,84,84,125 funded
NY	NY: 3701 loans \$4,20,77,050 funded
TX	TX: 2664 loans \$3,12,36,650 funded
FL	FL: 2773 loans \$3,00,46,125 funded
NJ	NJ: 1822 loans \$2,16,57,475 funded

4) IFERROR-SAFE RATIO — RECEIVED / FUNDED BY EMPLOYMENT LENGTH

Employment Length	Funded Amount	Received Amount	Received/Funded Ratio
< 1 year	\$0	\$0	n/a
1 year	\$3,28,83,125	\$3,54,98,348	1.08
2 years	\$4,49,67,975	\$4,92,06,961	1.09
3 years	\$4,39,37,850	\$4,75,51,832	1.08
4 years	\$3,76,00,375	\$4,09,64,850	1.09
5 years	\$3,69,73,625	\$4,03,97,571	1.09
6 years	\$2,56,12,650	\$2,79,08,658	1.09
7 years	\$2,08,11,725	\$2,25,84,136	1.09
8 years	\$1,75,58,950	\$1,90,25,777	1.08
9 years	\$1,50,84,225	\$1,65,16,173	1.09
10+ years	\$11,61,15,950	\$12,58,71,616	1.08